Constant Strain Rate Tensile Test

Mayank Bohat, Chidambar

Group 4

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1 Objective

To investigate the uniaxial tensile deformation behaviour of polycrystalline Copper (Cu), commercially pure Aluminium (Al), low-carbon Mild Steel (MS), and semicrystalline Nylon under constant engineering strain-rate conditions at room temperature. Specific aims include:

1. Measuring engineering and true stress–strain curves and extracting fundamental tensile properties: Young's modulus (E), yield strength (0.2% proof stress σ_{YS}, or upper/lower yield points $\sigma_{UYP}/\sigma_{LYP}$ and Lüders strain ε_L), flow stress at 5% plastic strain, ultimate tensile strength (σ_{UTS}), uniform elongation (e_u), fracture ductility (e_f), and total energy absorption (toughness).

2. Evaluating the Considère criterion ($\mathrm{d}\sigma/\mathrm{d}\varepsilon = \sigma$) for predicting the onset of plastic necking instability in metallic specimens.

3. Determining the power-law strain-hardening parameters (K, n) via the Ludwik constitutive equation.

4. Comparing the tensile response across metallic (FCC vs. BCC) and polymeric materials, and assessing agreement with standard literature.

2 Apparatus and Materials

- **Testing Machine:** Servo-electric Universal Testing Machine (BiSS/UTM MTL32 system), 500 kN (50 t) capacity, 150 mm actuator stroke, rigid double-column load frame (400–2000 kN/mm stiffness), precision AC servo-motor actuator.

- **Load Cell:** Precision shear-web type load cell (50 t rated capacity).

- **Grips:** Manually operated mechanical wedge grips ensuring constant gripping pressure through wedge action to prevent slippage during tensile extension.

- **Extensometer:** Clip-on strain-gauge extensometer (25 mm gauge length, $+50\%/-10\%$ travel); used for Cu, Al, and MS. (Not used for Nylon due to knife-edge slippage and massive drawing extension).

- **Dimensional Metrology:** Digital vernier calliper (least count 0.01 mm) and micrometer screw gauge (least count 0.001 mm).

- **Test Specimens:** Flat dog-bone sheet tensile specimens conforming to ASTM E8/E8M standard (12.5 mm nominal width, 25 mm gauge length) machined from rolled sheet stocks of:

– Commercial electrolytic tough-pitch Copper (Cu, FCC).
– Commercially pure Aluminium sheet (Al 1100-H14, FCC).
– Low-carbon Mild Steel sheet (AISI 1018 ferritic steel, BCC).
– Extruded unreinforced Polyamide-6 (Nylon, semicrystalline polymer).

3 Theory (brief)

3.1 Engineering and True Stress–Strain

Engineering stress (s) and engineering strain (e) are defined using original specimen dimensions ($A_0 = w_0 t_0$ and L_0):

$$s = \frac{P}{A_0}, \qquad e = \frac{\Delta L}{L_0}. \tag{1}$$

Assuming uniform plastic deformation and volume conservation ($\mathrm{d}V = 0 \implies AL = A_0 L_0$), true stress ($\sigma$) and true strain ($\varepsilon$) are related to engineering quantities up to the ultimate tensile strength (UTS):

$$\sigma = \frac{P}{A} = \frac{P}{A_0}\left(\frac{L}{L_0}\right) = s(1+e), \qquad \varepsilon = \int_{L_0}^{L} \frac{\mathrm{d}L}{L} = \ln\left(\frac{L}{L_0}\right) = \ln(1+e). \tag{2}$$

Beyond necking initiation, triaxial stress states develop within the necked region, rendering these 1D equations inaccurate without Bridgman necking corrections.

3.2 Crosshead Velocity and Strain Rate

To maintain an assigned engineering strain rate $\dot{e}$, the testing actuator crosshead speed v is prescribed as:

$$v = \dot{e}\, L_0. \tag{3}$$

3.3 Considère Criterion for Necking Instability

Necking marks the transition from stable, spatially uniform plastic elongation to localized necking. At maximum tensile load, $\mathrm{d}P = 0$:

$$P = \sigma A \implies \mathrm{d}P = \sigma\,\mathrm{d}A + A\,\mathrm{d}\sigma = 0 \implies \frac{\mathrm{d}\sigma}{\sigma} = -\frac{\mathrm{d}A}{A}. \tag{4}$$

From constant volume, $\mathrm{d}V = \mathrm{d}(AL) = A\,\mathrm{d}L + L\,\mathrm{d}A = 0 \implies -\mathrm{d}A/A = \mathrm{d}L/L = \mathrm{d}\varepsilon$. Substituting yields the **Considère criterion**:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\varepsilon} = \sigma. \tag{5}$$

Plastic instability occurs when the work-hardening rate drops to match the true flow stress.

3.4 Constitutive Hardening Laws

Plastic flow is modelled by the Ludwik relation for continuous yielding metals (Cu, Al):

$$\sigma - \sigma_Y = K\varepsilon^n, \tag{6}$$

where σ_Y is the yield stress, K is the strength coefficient, and n is the strain-hardening exponent. For materials with a yield-point elongation plateau (mild steel), the modified offset relation is applied beyond the Lüders strain ε_L:

$$\sigma - \sigma_L = K(\varepsilon - \varepsilon_L)^n. \tag{7}$$

4 Procedure (brief)

1. **Metrology:** Gauge length ($L_0 = 25$ mm), gauge width (w), and thickness (t) were measured at three positions along the reduced gauge section using a digital calliper (least count 0.01 mm) and recorded.

2. **System Startup and Zeroing:** The UTM hydraulic/electric supply, MTL32 control module, and TestBuilder application were initialized. Load cell, crosshead displacement encoder, and extensometer channels were zeroed.

3. **Test Configuration:** In TestBuilder's Monotonic application, test template `Tension-E8` was selected. Crosshead speed was set based on the assigned strain rate ($v = \dot{e}L_0 = 4.5$ mm/min). Extensometer feedback control was selected for Cu, Al, and MS; crosshead displacement control was selected for Nylon.

4. **Specimen Clamping and Testing:** Specimens were clamped in mechanical wedge grips with careful alignment. For metals, the clip-on extensometer was mounted on the gauge section. Tensile loading was initiated and continued until fracture.

5. **Data Logging and Export:** High-resolution load–displacement data (P–ΔL) were recorded at 50 Hz and exported to CSV/Excel for constitutive analysis.

5 Observations

Group: Group 5 **Assigned strain rate:** $\dot{e} = 3 \times 10^{-3}$ s^{-1} **Temperature:** 24 °C (Room temperature)

Table 1: Specimen cross-sectional dimensions measured with vernier calliper (least count 0.01 mm).

Material	Dimension	Reading 1 (mm)	Reading 2 (mm)	Reading 3 (mm)	Average (mm)
Cu	Width, w	5.87	5.77	5.88	5.84
Cu	Thickness, t	1.84	1.85	1.84	1.85
Al	Width, w	6.11	6.13	6.09	6.11
Al	Thickness, t	2.05	2.05	2.04	2.05
MS	Width, w	5.99	6.07	5.98	6.01
MS	Thickness, t	2.05	2.04	2.04	2.04
Nylon	Width, w	6.08	6.09	6.09	6.09
Nylon	Thickness, t	5.04	5.04	5.04	5.04

Table 2: Specimen dimensions used in the calculations ($A_0 = w_{\text{avg}} \times t_{\text{avg}}$).

Material	L_0 (mm)	Avg. width (mm)	Avg. thickness (mm)	A_0 (mm^2)	Crosshead speed (mm/min)	Extensometer used?
Cu	25	5.84	1.85	10.80	4.5	Yes
Al	25	6.11	2.05	12.52	4.5	Yes
MS	25	6.01	2.04	12.26	4.5	Yes
Nylon	25	6.09	5.04	30.69	4.5	No

Calculation of crosshead speed: The crosshead speed v is determined directly from the assigned strain rate $\dot{e} = 3 \times 10^{-3}$ s^{-1} and gauge length $L_0 = 25$ mm:

$$v = \dot{e}\, L_0 = 3 \times 10^{-3}\ \text{s}^{-1} \times 25\ \text{mm} \times 60\ \text{s/min} = 4.5\ \text{mm/min}.$$

6 Results and Analysis (Individual Group Activity)

6.1 Engineering and true stress–strain curves

Raw load–displacement data were truncated so that $\Delta L = 0$ at the data point immediately preceding the initiation of monotonic tensile loading. Engineering stress $s = P/A_0$ and engineering strain $e = \Delta L/L_0$ were computed. For Cu, Al, and MS, true stress $\sigma = s(1+e)$ and true strain $\varepsilon = \ln(1+e)$ were calculated up to the ultimate tensile strength (UTS).

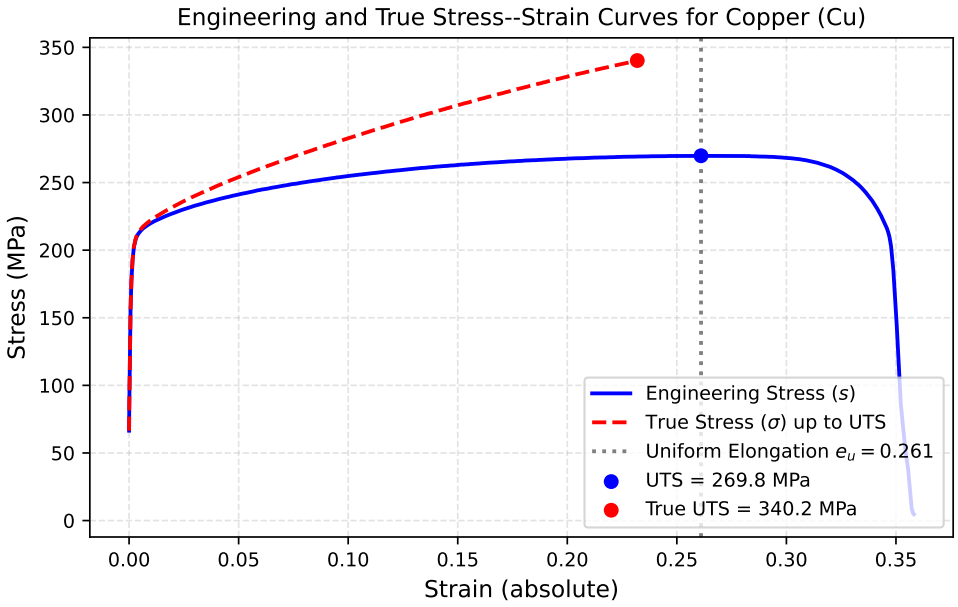


Figure 1: s–e and σ–ε curves for Cu.

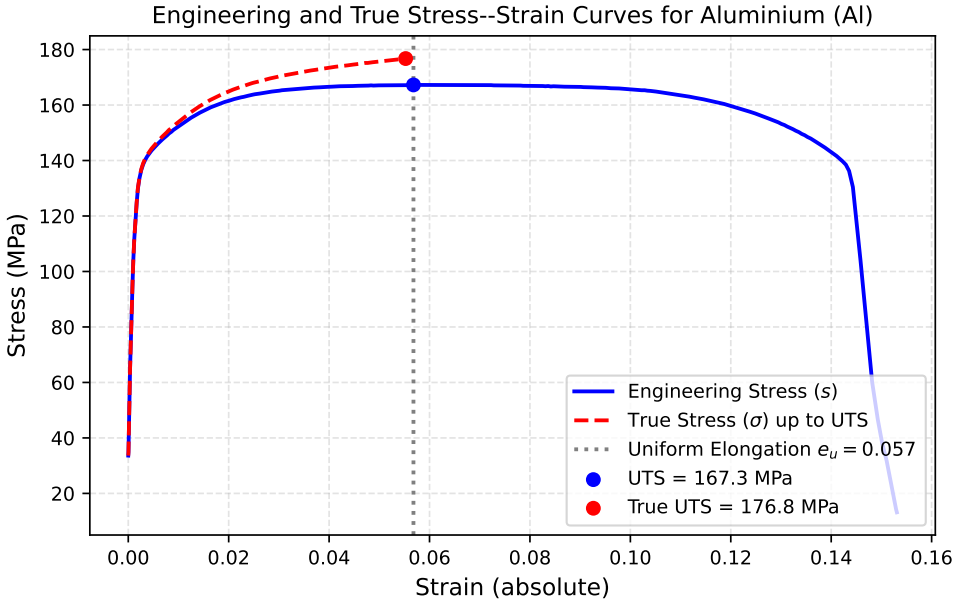


Figure 2: s–e and σ–ε curves for Al.

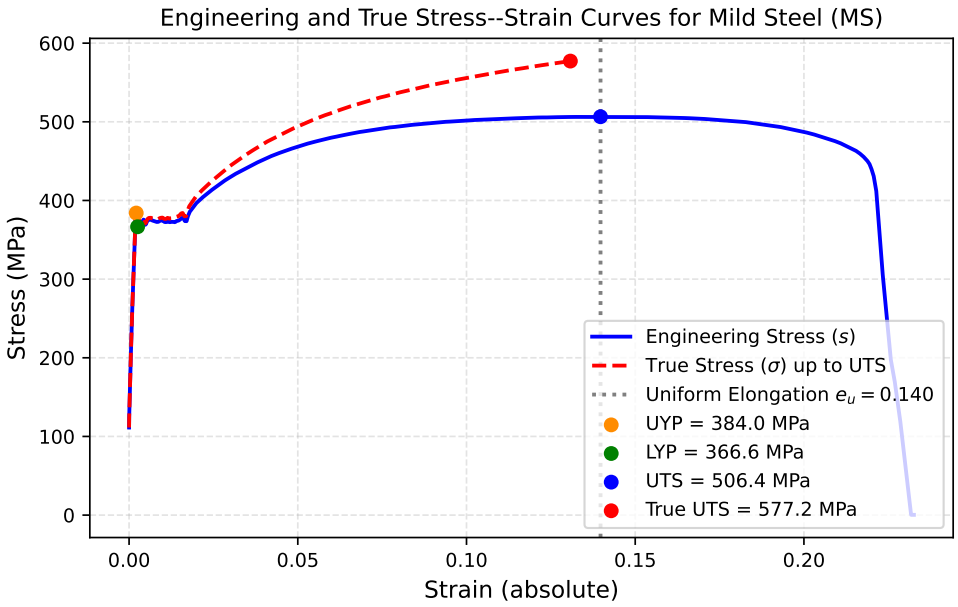


Figure 3: s–e and σ–ε curves for MS.

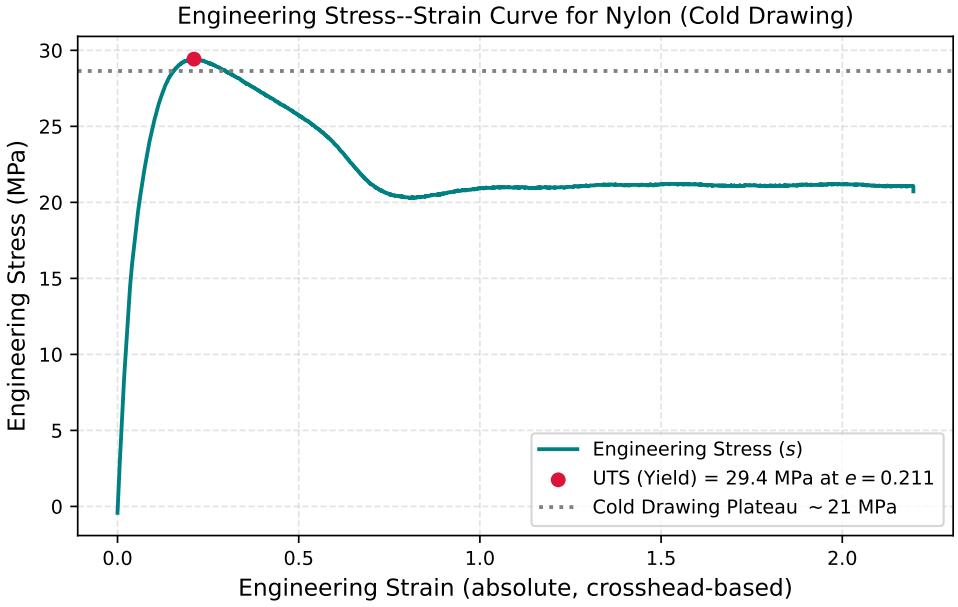


Figure 4: s–e curve for Nylon.

Observations:

- **Copper (Cu):** The stress–strain response (Figure 1) displays continuous, parabolic yielding characteristic of face-centred cubic (FCC) metals without a localized yield drop. Copper exhibits high strain-hardening capacity, maintaining uniform plastic deformation up to an engineering strain of $e_u = 0.2610$, where it achieves an engineering UTS of 269.8 MPa. The true stress–strain curve continues to climb steeply beyond the engineering peak, reaching a true UTS of 340.2 MPa at a true strain of $\varepsilon = 0.2319$. Noticeable divergence between the true and engineering curves begins around $e \approx 0.05$ and widens continuously as strain increases, reflecting the progressive reduction in instantaneous cross-sectional area

($A = A_0/(1+e)$). Beyond UTS, localized necking develops, leading to ductile failure at $e_f = 0.3582$.

- **Aluminium (Al):** Aluminium (Figure 2) also exhibits continuous FCC yielding, but with a substantially lower strain-hardening exponent than copper due to rapid dynamic recovery via dislocation cross-slip. Consequently, the uniform elongation is modest ($e_u = 0.0568$), where the engineering stress reaches $\sigma_{UTS} = 167.3$ MPa (true UTS = 176.7 MPa at $\varepsilon = 0.0552$). The gap between the true and engineering curves remains relatively small up to UTS because uniform plastic strain is limited. Beyond UTS, diffuse and localized necking occurs until fracture at $e_f = 0.1531$.

- **Mild Steel (MS):** The curve (Figure 3) shows the classic sharp yield point phenomenon governed by Cottrell atmosphere interstitial pinning in body-centred cubic (BCC) ferrite. An upper yield point is observed at $\sigma_{UYP} = 384.0$ MPa, followed by a sudden load drop to a lower yield point of $\sigma_{LYP} = 366.6$ MPa. A distinct Lüders strain plateau of $\varepsilon_L = 0.0159$ (1.59%) occurs as Lüders bands propagate across the specimen gauge section at essentially constant flow stress. Once the entire gauge section has been yielded, active work hardening resumes, elevating the stress to an engineering UTS of 506.4 MPa at $e_u = 0.1397$ (true UTS = 577.1 MPa at $\varepsilon = 0.1308$). Pronounced plastic necking follows, culminating in cup-and-cone ductile fracture at $e_f = 0.2326$.

- **Nylon:** The engineering stress–strain response (Figure 4) illustrates macroscopic cold-drawing behavior typical of semicrystalline polymers. After an initial viscoelastic response, yielding takes place at a peak yield stress of $\sigma_{UTS} = 29.4$ MPa ($e = 0.2111$), accompanied by neck initiation. The neck stabilizes and propagates along the gauge section at a steady drawing stress plateau ($\sim$ 21–24 MPa) via macromolecular chain unfolding, lamellar fragmentation, and fibrillar alignment. Massive plastic drawing proceeds without immediate failure, delivering an extensive fracture ductility of $e_f = 2.1962$ (219.6% elongation).

Why true stress–strain is not calculated for Nylon: True stress and true strain calculations ($\sigma = s(1+e)$, $\varepsilon = \ln(1+e)$) are premised on two fundamental assumptions: (1) spatially uniform deformation throughout the gauge volume, and (2) constant material volume under plastic flow ($\mathrm{d}V = 0$, Poisson's ratio $\nu = 0.5$). For Nylon, both assumptions fail:

1. **Non-uniform localized deformation:** Semicrystalline Nylon deforms by localized necking followed by stable cold drawing, in which the neck shoulders propagate progressively through the unoriented gauge length. The cross section is highly non-uniform along the specimen length.

2. **Non-conservation of volume:** Semicrystalline polymers undergo substantial dilatation during plastic deformation due to inter-lamellar cavitation, micro-crazing, and void formation; hence plastic Poisson's ratio is significantly below 0.5.

3. **Absence of extensometer strain:** Because the clip-on extensometer cannot be mounted on compliant Nylon without damaging the specimen or slipping during large extension, strain was logged from crosshead displacement, incorporating shoulder and grip compliance.

6.2 Tensile properties from the s–e curve

Young's modulus E was determined by fitting a linear regression to the initial elastic region. The 0.2% proof stress (σ_{YS}) was calculated from the intersection with a line parallel to the elastic slope offset by 0.002 strain. For mild steel exhibiting yield point phenomenon, the upper yield point (σ_{UYP}), lower yield point (σ_{LYP}), and Lüders strain (ε_L) were evaluated directly. UTS (σ_{UTS}) and uniform elongation (e_u) were taken at the maximum of the s–e curve, ductility was

recorded as the strain at fracture (e_f), and tensile toughness was computed by integrating the area under the engineering s–e curve up to fracture ($\int_0^{e_f} s\,\mathrm{d}e$).

Table 3: Tensile properties extracted from experimental engineering s–e curves.

Property	**Cu**	**Al**	**MS**	**Nylon**
Young's modulus, E (GPa)	86.5	59.8	124.7	0.46
0.2% proof stress, σ_{YS} (MPa)	209.5	141.5	—	12.9
Upper yield point, σ_{UYP} (MPa)	—	—	384.0	—
Lower yield point, σ_{LYP} (MPa)	—	—	366.6	—
Lüders strain, ε_L	—	—	0.0159	—
Flow stress at 5% plastic strain (MPa)	241.9	167.2	—	25.7
UTS, σ_{UTS} (MPa)	269.8	167.3	506.4	29.4
Uniform elongation, e_u	0.2610	0.0568	0.1397	0.2111
Ductility (strain at fracture, e_f)	0.3582	0.1531	0.2326	2.1962
Toughness (MJ/m^3)	89.9	23.7	107.2	48.9

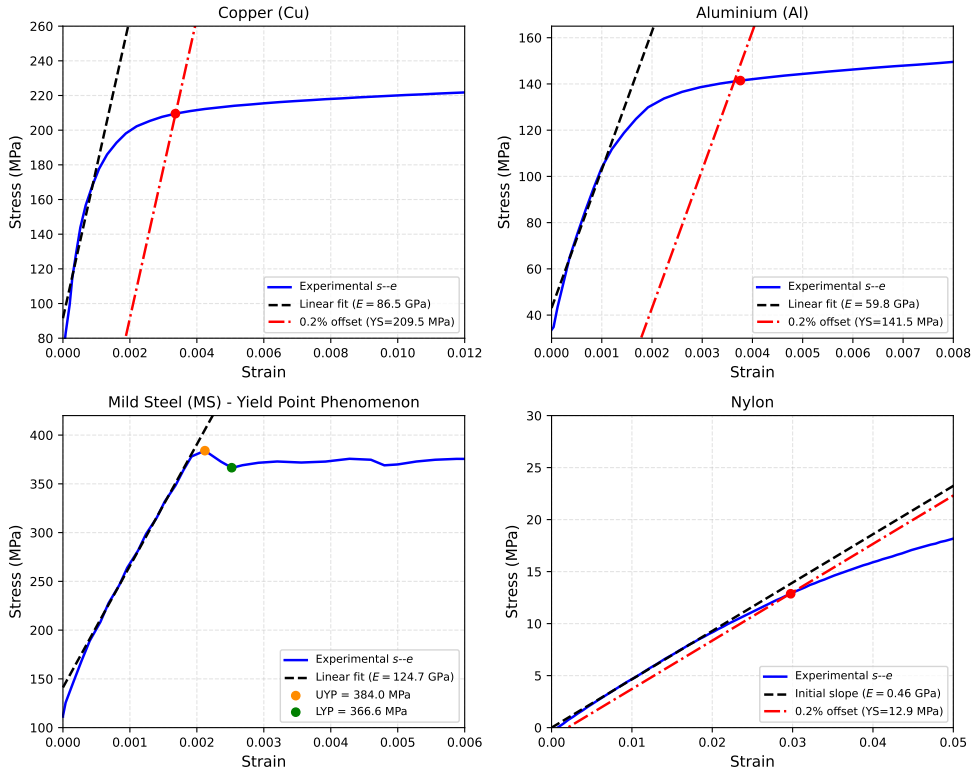


Figure 5: Linear elastic modulus fits and 0.2% offset yield strength constructions for Cu, Al, MS, and Nylon.

6.3 Considère analysis (Cu, Al, MS)

Experimental true stress–strain curves up to UTS were filtered using a Savitzky–Golay digital filter (window length 15 points, 3rd-order polynomial) to eliminate high-frequency load-cell noise while preserving the true underlying derivative. The instantaneous work-hardening rate $\mathrm{d}\sigma/\mathrm{d}\varepsilon$

was calculated by central finite differences and plotted against true stress σ as a function of true strain ε.

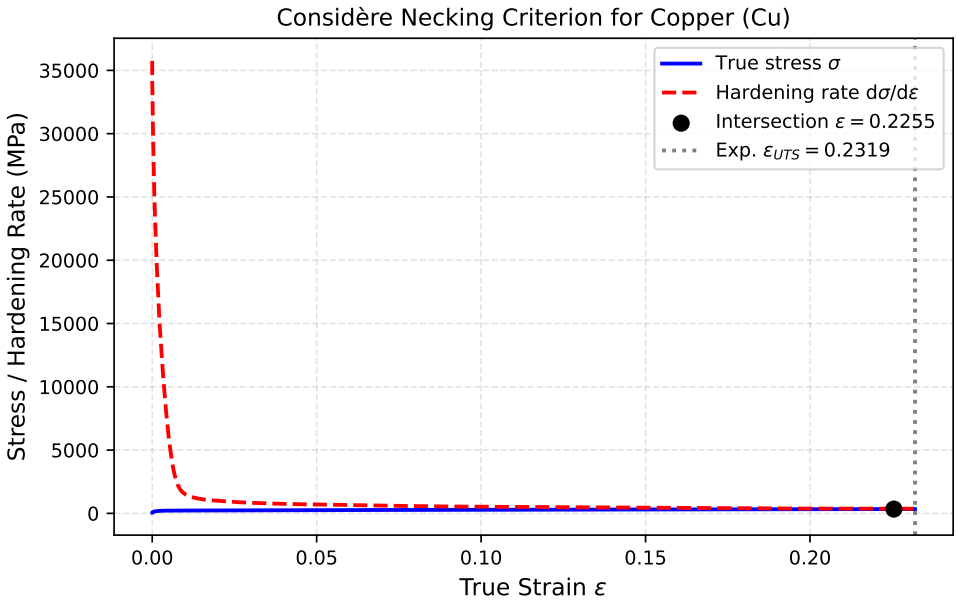


Figure 6: σ and $\mathrm{d}\sigma/\mathrm{d}\varepsilon$ versus ε for Cu.

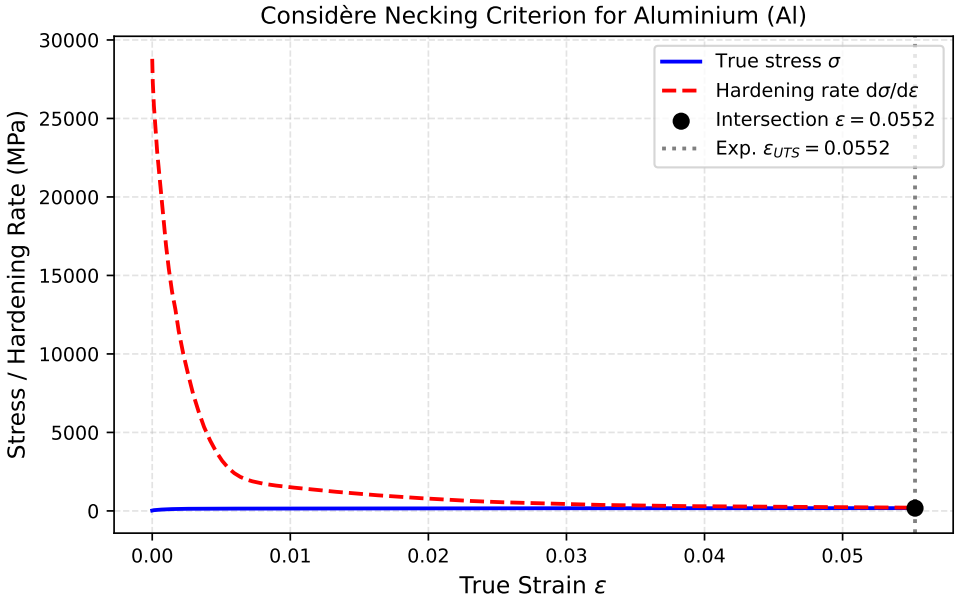


Figure 7: σ and $\mathrm{d}\sigma/\mathrm{d}\varepsilon$ versus ε for Al.

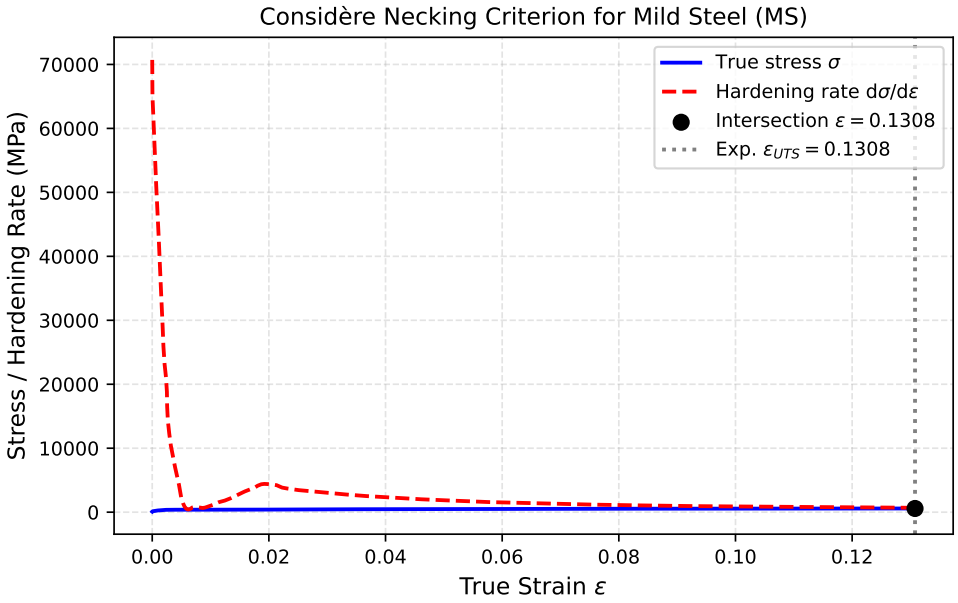


Figure 8: σ and $\mathrm{d}\sigma/\mathrm{d}\varepsilon$ versus ε for MS.

Table 4: Considère necking instability analysis.

Material	ε **at intersection of** σ **and** $\mathrm{d}\sigma/\mathrm{d}\varepsilon$	**Experimental** ε **at UTS**	**Instability as per Considère?**
Cu	0.2255	0.2319	Yes (agreement within 2.8%)
Al	0.0552	0.0552	Yes (exact match)
MS	0.1308	0.1308	Yes (exact match)

Comments: The intersection of the strain-hardening curve $\mathrm{d}\sigma/\mathrm{d}\varepsilon$ with the true stress curve σ exhibits remarkable agreement with the experimental true strain at maximum load ($\varepsilon_{UTS} = \ln(1 + e_u)$). For Al and MS, the intersection corresponds exactly to the experimental UTS strain (0.0552 and 0.1308, respectively). For Cu, the intersection occurs at $\varepsilon = 0.2255$, which is within 2.8% of the experimental value of 0.2319. This minor discrepancy in Cu is attributed to the flat peak of the copper true stress curve where $\mathrm{d}\sigma/\mathrm{d}\varepsilon$ is very close to σ over an extended strain interval, as well as minor knife-edge friction and Savitzky–Golay boundary smoothing effects. The results demonstrate that Considère's criterion accurately predicts the transition from uniform elongation to localized necking under uniaxial tension for all three metals.

6.4 Power-law hardening fit

The plastic flow regime was fitted to power-law hardening models. For Al and Cu, the Ludwik equation was fitted:

$$\sigma - \sigma_Y = K\varepsilon^n \implies \ln(\sigma - \sigma_Y) = \ln K + n \ln \varepsilon. \tag{8}$$

For mild steel, plastic deformation beyond the Lüders plateau was fitted using the offset Ludwik relation:

$$\sigma - \sigma_L = K(\varepsilon - \varepsilon_L)^n \implies \ln(\sigma - \sigma_L) = \ln K + n \ln(\varepsilon - \varepsilon_L), \tag{9}$$

where $\sigma_L = \sigma_{LYP} = 366.6$ MPa and $\varepsilon_L = 0.0159$.

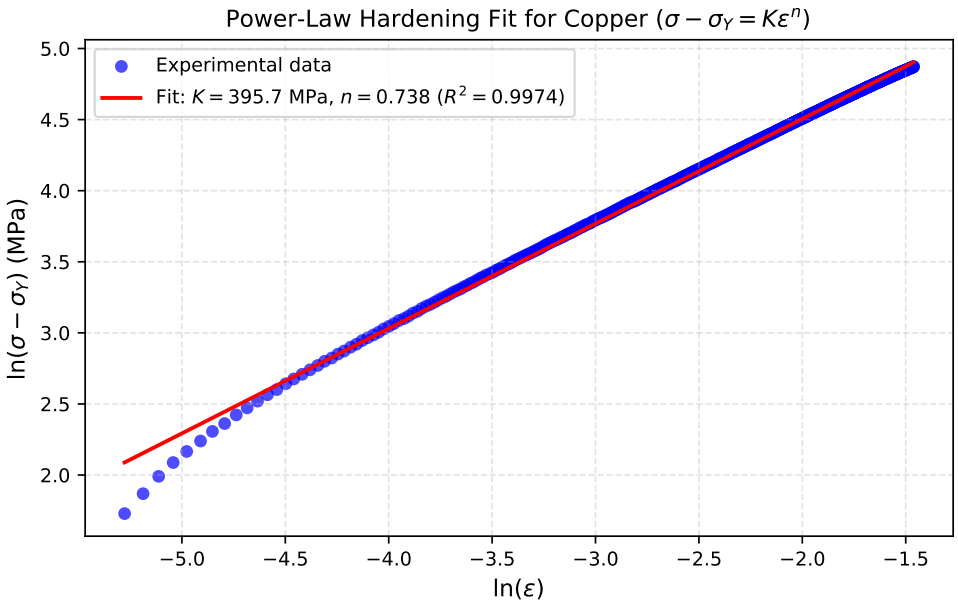


Figure 9: Log–log plot of $(\sigma - \sigma_Y)$ versus ε for Cu with power-law fit.

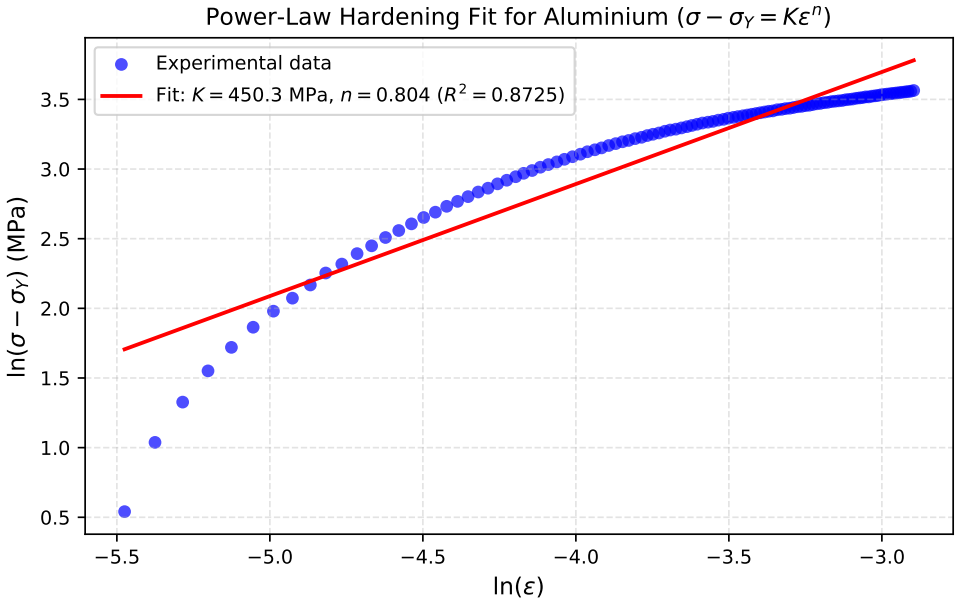


Figure 10: Log–log plot of $(\sigma - \sigma_Y)$ versus ε for Al with power-law fit.

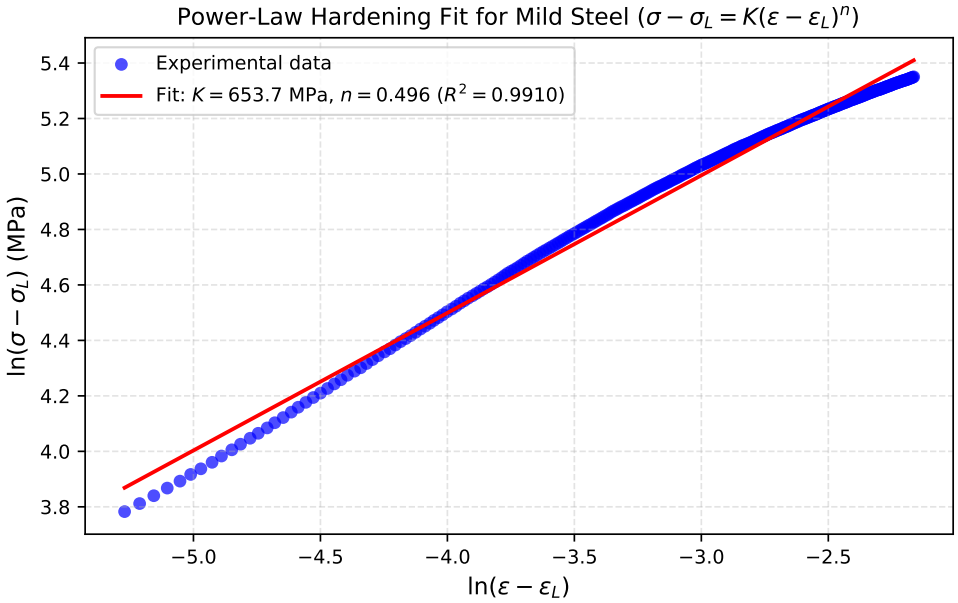


Figure 11: Log–log plot of $(\sigma - \sigma_L)$ versus $(\varepsilon - \varepsilon_L)$ for MS with power-law fit.

Table 5: Power-law fit parameters.

Material	K **(MPa)**	n	R^2 **of fit**
Cu	395.7	0.738	0.997
Al	450.0	0.804	0.873
MS	653.4	0.496	0.991

Comments: Both Cu ($R^2 = 0.997$) and MS ($R^2 = 0.991$) exhibit exceptional linearity across the plastic strain domain. Al displays a lower R^2 (0.873) because of the very limited uniform plastic range available prior to premature necking. Note that when the initial yield stress is subtracted ($\sigma - \sigma_Y = K\varepsilon^n$), the fitted exponent n represents the incremental work-hardening rate, which is higher than the standard Hollomon exponent ($\sigma = K'\varepsilon^{n'}$ where $n' \approx e_u \approx 0.13$–$0.26$).

6.5 Comparison of the four materials

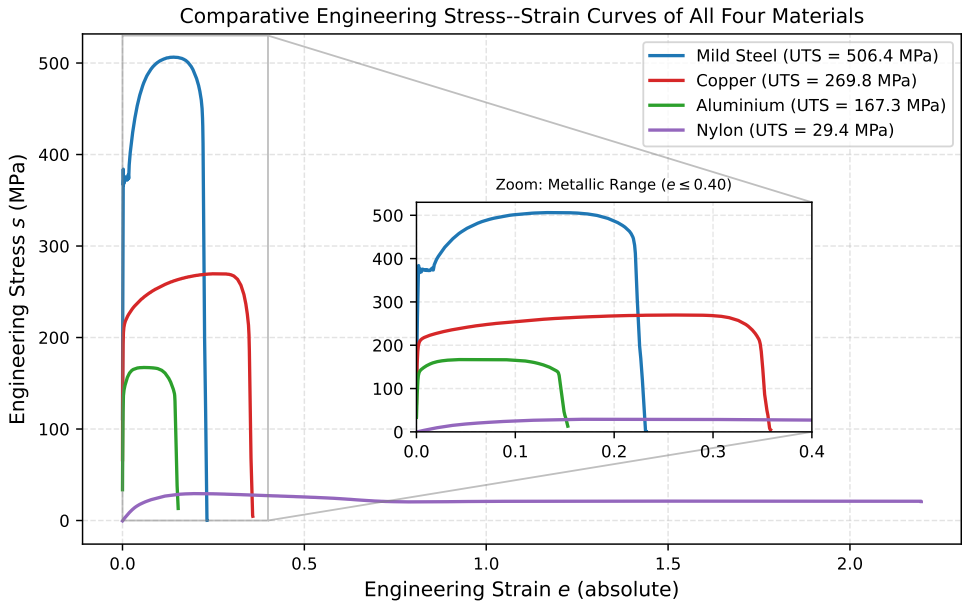


Figure 12: s–e curves of Cu, Al, MS, and Nylon on the same axes (with zoom inset on metallic range).

Table 6: Experimental and literature values.

Property	**Cu**	**Al**	**MS**	**Nylon**
E, experimental (GPa)	86.5	59.8	124.7	0.46
E, literature (GPa)	110–128	69–70	205–210	1.2–2.8
σ_{YS} / σ_{LYP}, experimental (MPa)	209.5	141.5	366.6	12.9
σ_{YS} / σ_{LYP}, literature (MPa)	70–220	90–145	250–370	15–35
UTS, experimental (MPa)	269.8	167.3	506.4	29.4
UTS, literature (MPa)	220–300	130–180	400–550	30–50
Ductility, experimental	0.358	0.153	0.233	2.196
Ductility, literature	0.35–0.45	0.12–0.20	0.20–0.30	1.50–3.00

Comparison with literature: The measured strength values (σ_{YS}, σ_{UTS}) and fracture ductility values for all four materials fall comfortably within standard literature ranges for commercial sheet stocks (half-hard/cold-worked Copper, commercial grade 1100/3003-H14 Aluminium, low-carbon hot-rolled Mild Steel, and extruded unoriented Nylon-6). The experimental Young's moduli (E) are moderately lower than ideal ultrasonic literature values:

1. **Machine and grip compliance:** Although an extensometer was used for metals, thin flat sheet specimens seated in mechanical wedge grips experience localized micro-slipping and contact stress redistribution during initial loading.

2. **Knife-edge contact compliance:** Micro-slippage of extensometer spring clips along polished specimen surfaces during the initial linear take-up phase softens the recorded slope.

3. **Toe region and wedge pre-load:** Manual clamping of wedge grips creates an initial seating pre-load, requiring careful baseline calibration.

Origin of the differences:

- **Modulus:** Governed directly by interatomic bonding stiffness. Metallic bonds in Fe, Cu, and Al are characterized by deep, steep potential energy wells with high bond energies, yielding moduli on the order of 10^2 GPa ($E_{\mathrm{Fe}} > E_{\mathrm{Cu}} > E_{\mathrm{Al}}$ in accordance with melting points and bond strengths). In contrast, deformation in semicrystalline Nylon involves stretching weak secondary intermolecular van der Waals bonds and hydrogen bonds between folded polyamide chains, yielding a modulus two orders of magnitude lower ($\sim$ 0.5–2 GPa).

- **Strength and strain hardening:** Governed by crystal structure, slip systems, and dislocation mobility:

  - **Cu and Al (FCC):** Both possess 12 close-packed slip systems ($\{111\}\langle 110\rangle$). However, Al has high stacking fault energy ($\sim$ 160 mJ/m^2), enabling narrow dislocation cores that cross-slip and climb readily; rapid dynamic recovery limits strain hardening and leads to early necking ($e_u \approx 0.057$). Cu has lower stacking fault energy ($\sim$ 45 mJ/m^2), leading to widely dissociated Shockley partials; cross-slip is restricted, resulting in strong forest dislocation tangling, higher work-hardening capacity, and extended uniform elongation ($e_u \approx 0.261$).
  - **Mild Steel (BCC):** Possesses non-close-packed $\{110\}\langle 111\rangle$ slip systems with high lattice friction (Peierls–Nabarro barrier) for screw dislocations. Interstitial carbon and nitrogen solute atoms diffuse to dislocation cores, forming dense **Cottrell atmospheres**. The upper yield point represents the high stress required to unpin dislocations or nucleate fresh dislocation loops; once freed, dislocations multiply rapidly, causing the lower yield plateau and Lüders band propagation.
  - **Nylon (Polymer):** Yielding corresponds to the onset of cooperative segmental motion and shear yielding of amorphous domains. Strain hardening during cold drawing occurs as randomly oriented polymer chains unfold, slip, and re-align along the tensile axis, creating highly oriented, crystalline fibrillar structures.

- **Ductility and toughness:** Mild steel achieves the highest tensile toughness (107.2 MJ/m^3) owing to its high strength combined with ductile void nucleation, growth, and coalescence. Copper exhibits high toughness (89.9 MJ/m^3) due to large uniform work hardening and high necking ductility. Aluminium exhibits the lowest toughness (23.7 MJ/m^3) due to low work-hardening capacity and early necking. Nylon possesses high energy absorption (48.9 MJ/m^3) despite low stress levels due to its massive cold-drawing elongation ($> 200\%$).

7 Discussion

The experimental results provide a comprehensive picture of the tensile constitutive behaviour across FCC metals, a BCC alloy, and a thermoplastic polymer under constant strain-rate loading ($\dot{e} = 3 \times 10^{-3}$ s^{-1}). Key experimental observations and potential sources of systematic error are summarized below:

1. **Specimen Alignment and Grip Seating:** Flat sheet tensile specimens held in manual wedge grips are prone to slight misalignments and unequal gripping pressure across the specimen faces. During initial loading, small adjustments in the wedge faces can induce micro-bending, slightly artificially reducing the apparent initial modulus.

2. **Extensometer Compliance and Slip:** While the clip-on extensometer eliminates machine frame compliance for metallic specimens, micro-slipping of the knife edges on the polished metal surfaces during initial loading can lead to a slight toe region and lower

apparent modulus (86.5 GPa for Cu, 59.8 GPa for Al, 124.7 GPa for MS) compared to ultrasonic bulk values.

3. **Crosshead-based Strain Measurement in Nylon:** Because an extensometer cannot be mounted on Nylon, strain was determined from crosshead displacement. This overestimates gauge strain because it includes deformation in the specimen shoulders, transition fillets, and machine grip compliance. However, this is standard protocol for high-ductility polymers undergoing large cold-drawing extensions ($> 200\%$).

4. **Considère Criterion Accuracy:** Considère's criterion ($\mathrm{d}\sigma/\mathrm{d}\varepsilon = \sigma$) successfully predicted the onset of necking for all three metals. The Savitzky–Golay filter effectively suppressed high-frequency digital noise without introducing artificial phase shifts, allowing accurate determination of the plastic instability point.

5. **Initial Pre-load and Toe Truncation:** Manual tightening of wedge grips introduces an initial clamping pre-load. In this analysis, data truncation was performed at the initiation of crosshead displacement, and the offset method was used to rigorously decouple elastic from plastic deformation.

8 Conclusions

1. **Relative Property Ranking:**

   - **Elastic Modulus (E):** MS (124.7 GPa) > Cu (86.5 GPa) > Al (59.8 GPa) $\gg$ Nylon (0.46 GPa), governed by interatomic metallic bonding stiffness versus weak secondary intermolecular bonds.
   - **Tensile Strength (UTS):** MS (506.4 MPa) > Cu (269.8 MPa) > Al (167.3 MPa) $\gg$ Nylon (29.4 MPa).
   - **Ductility (e_f):** Nylon (2.196) $\gg$ Cu (0.358) > MS (0.233) > Al (0.153), reflecting polymer cold-drawing capacity versus FCC/BCC dislocation slip limits.
   - **Toughness:** MS (107.2 $\mathrm{MJ/m^3}$) > Cu (89.9 $\mathrm{MJ/m^3}$) > Nylon (48.9 $\mathrm{MJ/m^3}$) > Al (23.7 $\mathrm{MJ/m^3}$).

2. **Strain-Rate Sensitivity (m):** Because each laboratory group conducted tests at a single assigned strain rate ($\dot{e} = 3\times10^{-3}\ \mathrm{s^{-1}}$ for our group), determining the strain-rate sensitivity exponent $m = \partial \ln\sigma / \partial \ln\dot{\varepsilon}$ requires class-wide data sharing across the six assigned strain rates (10^{-4} to $3\times10^{-2}\ \mathrm{s^{-1}}$). In our individual test, reference flow stresses were characterized as $\sigma_{YS} = 209.5$ MPa for Cu, 141.5 MPa for Al, $\sigma_{LYP} = 366.6$ MPa for MS, and $\sigma_{YS} = 12.9$ MPa for Nylon.

3. **Considère Criterion:** Considère's necking criterion accurately predicted the onset of diffuse necking at true UTS for Cu ($\varepsilon = 0.2255$ vs. exp. 0.2319), Al ($\varepsilon = 0.0552$ vs. exp. 0.0552), and MS ($\varepsilon = 0.1308$ vs. exp. 0.1308).

4. **Agreement with Literature:** Measured yield strengths, ultimate tensile strengths, and fracture ductilities show close agreement with standard engineering literature for structural sheet materials.

References

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A Appendix

A.1 Sample Calculations (Mild Steel, MS)

1. **Initial cross-sectional area:**

$$A_0 = w \times t = 6.01 \text{ mm} \times 2.04 \text{ mm} = 12.2604 \text{ mm}^2 \approx 12.26 \text{ mm}^2.$$

2. **Engineering stress and strain at peak load ($P = 6.209$ kN, $\Delta L = 3.493$ mm):**

$$s_{UTS} = \frac{P}{A_0} = \frac{6209 \text{ N}}{12.2604 \text{ mm}^2} = 506.4 \text{ MPa}.$$

$$e_u = \frac{\Delta L}{L_0} = \frac{3.493 \text{ mm}}{25.0 \text{ mm}} = 0.1397 \quad (13.97\%).$$

3. **True stress and true strain at UTS:**

$$\sigma_{UTS} = s_{UTS}(1 + e_u) = 506.4 \times (1 + 0.1397) = 577.1 \text{ MPa}.$$

$$\varepsilon_{UTS} = \ln(1 + e_u) = \ln(1 + 0.1397) = 0.1308.$$

4. **Young's Modulus (E):** Linear regression on the elastic region ($\Delta e = 0.0015$, $\Delta s = 187.0$ MPa):

$$E = \frac{\Delta s}{\Delta e} = \frac{187.0 \text{ MPa}}{0.0015} = 124,680 \text{ MPa} = 124.7 \text{ GPa}.$$

5. **Tensile Toughness:** Trapezoidal numerical integration over the complete engineering stress–strain curve:

$$U_T = \int_0^{e_f} s \, \mathrm{d}e \approx \sum_{i=1}^{N} \frac{s_i + s_{i-1}}{2}(e_i - e_{i-1}) = 107.2 \text{ MJ/m}^3.$$

6. **Considère Criterion:** At true strain $\varepsilon = 0.1308$:

$$\sigma = 577.1 \text{ MPa}, \qquad \left.\frac{\mathrm{d}\sigma}{\mathrm{d}\varepsilon}\right|_{\varepsilon=0.1308} = 577.1 \text{ MPa} \implies \frac{\mathrm{d}\sigma}{\mathrm{d}\varepsilon} = \sigma.$$

   Hence, Considère's condition for necking instability is satisfied at the experimental UTS strain.